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Bhabani Shankar Dey

Assistant Professor (Research), School of Electrical Engineering, Vellore Institute of Technology, Vellore, India

About Me

I am currently working as an Assistant Professor (Research) in the School of Electrical Engineering, Vellore Institute of Technology, Vellore. Prior to that, I worked as a postdoctoral researcher with **Prof. Pushpak Jagtap** at Robert Bosch Centre for Cyber Physical Systems, **IISc Bangalore** after completing my Ph.D. in Electrical Engineering Department, **Indian Institute of Technology Delhi**. At IISc, I was working towards safe and stable human-robot collaboration, mitigating the Sim2Real Gap, and designing data-driven controllers for unknown systems. In my PhD, I broadly worked on the analysis design of switched control systems and their applications with **Prof. Indra Narayan Kar**. I considered switched systems as a benchmark problem, which has numerous engineering applications. Some of the challenges that I addressed are compositionality, xeno behaviour, and stability of large-scale interconnected systems.

Research Interests

Human-Robot Collaboration, Data-Driven Control Design for Unknown Systems, Control Design for Switched and Hybrid Systems, Sim2Real Gap, Medical Robotics

Experience

Assistant Professor (Research), School of Electrical Engineering June 2026 – Till Date
Vellore Institute of Technology, Vellore, India

Postdoctoral Researcher, Robert Bosch Centre for Cyber-Physical Systems, Indian Institute of Science (IISc) Bangalore October 2024 – May 2026

- Advisor: Prof. Pushpak Jagtap

An impedance control approach towards safe human–robot collaboration

Studied physical human–robot interaction where the robot has to remain compliant to the human and, at the same time, respect safety limits on interaction force and on the workspace. Instead of detuning the interaction dynamics to buy safety, safety was enforced by shaping the reference trajectory supplied to an admittance/impedance law, so that compliance is preserved while constraints are met. Contraction and incremental-stability arguments were used to certify convergence of the closed loop, and adaptive versions were derived for uncertain human and environment parameters. *Outputs: [J4], [C4], [C5].*

Neural network based formal verification and control of unknown robotic systems

Developed data-driven controllers for systems whose dynamics are not known analytically, together with certificates that can be checked rather than assumed. Neural controllers and neural Lyapunov-type certificates were trained from sampled trajectories, and their validity conditions were then verified over compact sets, giving incremental input-to-state stability guarantees for the learned closed loop. The framework was extended to discrete-time systems, to switched nonlinear systems, and to backstepping-style designs for cascaded structures. *Outputs: [J3], [R1], [R2], [R3], [C2], [C3].*

A control theoretical framework to minimise the Sim2Real gap

Addressed the loss of performance when a controller tuned in simulation is deployed on hardware. The mismatch between the simulator and the physical plant was treated as a bounded discrepancy, and incremental stability was used to bound how far the trajectories of the real system can drift from the simulated ones, so that policies transferred to hardware retain trajectory-level guarantees. *Output: [R4].*

Broader activity

Robotics Lab Setup for Centre of Excellence in VIT Vellore; Co-organiser of CyPhySS 2024 at IISc Bangalore; supervision and mentoring of Master's and PhD students on the projects above; invited seminars on human–robot collaboration and on incremental stability of switched systems.

Education

PhD, Control and Automation, Department of Electrical Engineering, Indian Institute of Technology Delhi July 2018 – October 2024

- Advisor: Prof. Indra Narayan Kar
- CGPA: 9.55/10.0

Thesis: Incremental Stability based Analysis and Design of Nonlinear Systems

We have employed incremental stability-based tools to design an observer/estimator and a controller for several classes of nonlinear systems, such as switched systems and singularly perturbed systems. Using Contraction theoritic tools, non-smooth dynamical systems were analysed, which represent many engineering systems. The thesis also targets large-scale complex interconnected switched systems and small gain characterisation to ensure stability of the overall architecture.

M.Tech., Control and Industrial Automation, Department of Electrical Engineering, National Institute of Technology Silchar August 2016 – July 2018

- Advisors: Prof. Binoy Krishna Roy and Prof. Manas Kumar Bera
- CGPA: 9.5/10.0, Special Award for highest marks in the M.Tech. batch.

Thesis: Control of Cancerous Tumour Growth by Chemotherapy

In this work, a three-dimensional cancer model is considered. The competition terms include tumour cells, healthy cells, and immune cells. The nature of the competition between the populations of tumour cells, healthy host cells, and immune cells results in chaotic behaviour. To control chaotic behaviour, some nonlinear control techniques have been adopted. There is also the effect of noise and perturbations. Moreover, the control input, which is a chemotherapeutic drug, should not affect the healthy cell populations. Incorporating all such physiological details, second-order supertwisting control has been designed.

B.Tech., Department of Electrical Engineering, C V Raman College of Engineering August 2009 – June 2013

- CGPA: 8.39/10.0

Thesis: Robust Control Design Using Optimisation Techniques

Coursework During PhD: Linear Control Theory, Optimal Control, Nonlinear Systems, Nonlinear Control, Adaptive Control, Robust Control, Stochastic Filtering, Estimation, Industrial Automation, Multi-Agent Systems, Industrial Instrumentation, Advanced Robotics, Human-Robot Collaboration, Discrete-Time Control System.

Journal Publications

[J1] **Dey, B.S.**, Kar, I.N. and Jagtap, P., 2026. On incremental stability of interconnected switched systems. *International Journal of Systems Science*, 57(4), pp.895-910.

[J2] **Dey, B.S., Paul**, C.K. and Kar, I.N., 2026. A matrix measure framework for incremental stability of mixed-mode switched systems. *International Journal of Systems Science*, pp.1-16.

[J3] Basu, A., **Dey, B.S.** and Jagtap, P., 2026. Neural controller for incremental stability of unknown continuous-time systems. *International Journal of Systems Science*, pp.1-16.

[J4] Paul, C.K., **Dey, B.S.** and Kar, I.N., 2026. Safety in Admittance Control Using Reference Trajectory Shaping. *International Journal of Human–Computer Interaction*, pp.1-17. **(Corresponding Author)**

[J5] Nath, A., Kumar, M. and **Dey, B.S.**, 2026. An Inverse Sinh-based Robust Discrete-Time Model Reference Sliding Mode Control for Systems with Unmatched Uncertainty. *Journal of the Franklin Institute*, p.109013. **(Corresponding Author)**

[J6] Bora, R.M., **Dey, B.S.** and Kar, I.N., 2026. 2-Contraction and Poincaré-Index Theory Based Framework for Stability Analysis of Nonlinear Dynamical Systems with Multiple Equilibria. *International Journal of Control, Automation, and Systems*, pp.1-15.

[J7] Rayguru, M.M., **Dey, B.S.** and Kar, I.N., 2021. Output feedback for contraction of nonlinear singularly perturbed systems. *International Journal of Control*, pp.1-12. **(Corresponding Author)**

[J8] Sangeerth, P., **Dey, B.S.** and Jagtap, P. Reducing Sim2Real Gap: An Incremental Stability Approach. (Accepted, IEEE Robotics and Automation Letters)**(Corresponding Author)**

[J9] **Dey, B.S.**, Bera, M.K. and Roy, B.K., 2018. Nonlinear active control of a cancerous tumour. *International Journal of Engineering & Technology*, 7(2.21), pp.72-76.

UNDER REVIEW

[R1] **Dey, B.S.**, Basu, A. and Jagtap, P., 2026. Input-to-State Stabilizing Neural Controllers for Unknown Switched Nonlinear Systems within Compact Sets. *arXiv preprint* arXiv:2601.14643. **(Under revision, Automatica)**

[R2] Basu, A., **Dey, B.S.** and Jagtap, P., 2025. Formally Verified Neural Network Controllers for Incremental Input-to-State Stability of Unknown Discrete-Time Systems. *arXiv preprint* arXiv:2503.04129. **(Under revision, IEEE Transactions on Systems, Man, and Cybernetics)**

[R3] Smith Sundarsingh, D., **Dey, B.S.** and Jagtap, P., 2024. Backstepping Design for Incremental Input-to-State Stabilization of Unknown Systems. *arXiv e-prints*, arXiv:2411. **(IEEE TAC)**

[R4] Paul, Chayan, **Dey, B.S.** and Kar, I.N.,Safe Physical Human-Robot Interaction via Fixed-Time Concurrent-Learning Impedance Control. **(Under revision, IEEE Transactions on Automation Science and Engineering)**

Conference Publications

[C1] **Dey, B.S.**, Kar, I.N. and Jagtap, P., 2025. Formal Guarantees for Incremental Stability in Piecewise Smooth Cascaded Switched Systems. In *2025 Eleventh Indian Control Conference (ICC)* (pp. 277-282). IEEE.

[C2] Sangeerth, P., Smith Sundarsingh, D., **Dey, B.S.** and Jagtap, P., 2025. Controller for Incremental Input-to-State Practical Stabilization of Partially Unknown Systems with Invariance Guarantees. In *2025 Eleventh Indian Control Conference (ICC)*, Bangalore, India (pp. 283-288). IEEE.

[C3] Basu, A., **Dey, B.S.** and Jagtap, P., 2025, June. Formally verified neural Lyapunov function for incremental input-to-state stability of unknown systems. In *2025 European Control Conference (ECC)* (pp. 446-451). IEEE.

[C4] Paul, C.K., **Dey, B.S.** and Kar, I.N., 2024, July. Safe Human Robot-Interaction using Switched Model Reference Admittance Control. In *CoDIT* (pp. 1589-1594). IEEE.

[C5] Paul, C.K., Patel, N., **Dey, B.S.** and Kar, I.N., 2024. Adaptive Impedance Control: A Contraction Theory Approach. *IFAC-PapersOnLine*, 57, pp.83-88.

[C6] **Dey, B.S.**, Banerjee, U. and Kar, I.N., 2023, July. Harmonics Suppression in a Class of Switched Mechanical System: A Contraction Theory Approach. In *2023 9th International Conference on Control, Decision and Information Technologies (CoDIT)* (pp. 1737-1742). IEEE.

[C7] **Dey, B.S.**, Nath, A., Patel, A. and Kar, I.N., 2022, December. A contraction theory approach to observer-based controller design for glucose regulation in type 1 diabetes. In *2022 Eighth Indian Control Conference (ICC)* (pp. 85-90). IEEE.

[C8] Banerjee, U., **Dey, B.S.**, Kar, I.N. and Saha, S.K., 2022, September. Robust Artificial Delay-based Impedance Control of Robotic Manipulators with Uncertain Dynamics. In *2022 16th International Workshop on Variable Structure Systems (VSS)* (pp. 89-94). IEEE.

[C9] **Dey, B.S.** and Kar, I.N., 2022. Switching Control for Euler-Lagrange Dynamics: An Incremental Stability Approach. *IFAC-PapersOnLine*, 55(1), pp.555-560.

[C10] **Dey, B.S.**, Bera, M.K. and Roy, B.K., 2018, July. Super twisting sliding mode control of cancer chemotherapy. In *2018 15th International Workshop on Variable Structure Systems (VSS)* (pp. 343-348). IEEE.

Invited Talks

- A talk on "Sim2Real Gap" at **Workshop on Control in Robotics: From Micro to Micro, IIT Kanpur**.
- A talk on "Human Robot Collaboration" at **ARML, NIT Patna**.
- A talk on "Control at Crossroad: Is Learning a Rescue?" in the **Department of Mechatronics, IIEST Shibpur**.
- A talk on "Incremental Stability Analysis of Switched System: A broad overview" in the online seminar series in **IISc Bangalore**.
- A talk on "Human Robot Collaboration" in the **ANRF symposium at IIEST Shibpur**.

Teaching Experience

Applied Linear and Nonlinear Control, Nonlinear Systems, Optimal Control, Basic Electrical, Contraction Theory and Applications.

Professional Service

Reviewer for: Automatica, IEEE TAC, IEEE Transactions on Mechatronics, IEEE CSS Letters, IEEE RA-L, International Journal of Control, Computer Methods and Programs in Biomedicine, Nonlinear Dynamics, IFAC World Congress, IEEE Conferences.

ORGANISATION AND COMMITTEES

- Member of Centre of Excellence for Robotics at VIT
- Associate Editor at ACODS 2026.
- Co-organiser at CyPhySS 2024, IISc Bangalore.
- Volunteer in Indian Control Conference, 2019.
- Co-Organiser in Control Colloquium, IIT Delhi.
- Publicity Chair, IEEE PES Student Chapter, NITS.

Awards and Achievements

- Received special medal for securing highest marks in M.Tech.
- Special Award in M.Tech for Batch Topper.
- Qualified Graduate Aptitude Test Examination, 2016.
- Fellowship during M.Tech.
- Fellowship during PhD.
- Research Scholar Travel Award to attend conference in Rome, Italy.
- EECI Overseas Grant to visit the International Graduate School of Control in Paris, France.
- Passed 'Business English Certificate Preliminary Council of Europe Level B1' from Cambridge University.
- Selected and participated in state-level Scout & Guide.
- Won the literary champion award in intermediate.

Training and Industrial Tours

- GIAN course on Sliding Mode Control: Robust Control Technique, NIT Silchar.
- GIAN course on Robustness, Fragility, Optimality and Modern PID Control, NIT Silchar.
- GIAN course on Numerical methods, NIT Silchar.
- 30 days of training on transmission and distribution in OPTCL.
- 15 days training on Industrial Automation (PLC) by Bosch Rexroth.
- Visited the captive power plant of Rourkela Steel Plant.
- Visited PGCIL Kaniha, NTPC Talcher and Rengali hydro project.

Technical Skills

Skills: MATLAB, Python, Model-based design, Simulink, Data Driven Cotrol, CARLA-Simulator